

Setting up docker container

```
docker run --name my-postgres -e POSTGRES_PASSWORD=mysecretpassword -d -p 5432:5432 postgres
```

Entering the PSQL shell

```
docker exec -it my-postgres psql -U postgres
```

Creating students table

```
-- 1. Create the table
CREATE TABLE grades (
    id INT,
    name VARCHAR(50),
    grade INT
);

-- 2. Insert 1 million rows
INSERT INTO grades (id, name, grade)
SELECT
    x,
    'Student_' || x,
    floor(random() * 101)
FROM generate_series(1, 5000000) AS x;

-- 3. Create indexes on both 'id' and 'grade'
CREATE INDEX idx_students_id ON grades(id);
CREATE INDEX idx_students_grade ON grades(grade);
```

Number of pages

```
SELECT relpages
FROM pg_class
WHERE relname = 'students';
```

Commands are gonna exec

```
--1
explain select * from students
--2
explain select * from students where id = 10;
```

Practical cost for seq scan WITHOUT an index

$\text{Cost} = (\text{cpu_tuple_cost} * \text{rows}) + (\text{pages})$